# EXPERIMENT 1 0

# PerformanceOptimizationinZoho Creator

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## Aim

To demonstrate performance optimization techniques in a Zoho Creator application, particularly efficient record retrieval and workflow processing.

## Introduction

Performance optimization in low-code applications means designing forms, workflows and scripts so that they use data and processing only as far as needed. Although a low-code platform handles much of the underlying work, the way a workflow is written still influences how efficiently it runs.

Retrieving more records than required, or repeatedly processing records that are not relevant to the task, causes unnecessary data retrieval and processing. In a small application this may not be noticeable, but as the number of records and users grows it can reduce application efficiency. Efficient workflows in Zoho Creator therefore retrieve only the data they need and avoid unnecessary processing, which supports scalable and maintainable applications.

## Application Used

The practical was performed in Zoho Creator using the Job Application Form application. Deluge scripting was used to write the logic of a form workflow that sends an email notification to the applicant. The Zoho Creator account shows an upgrade option in the application header, so some advanced features may require an upgrade; no upgrade-only feature is claimed as enabled or configured in this record.

## Performance Optimization Concept

The key optimization demonstrated is **criteria-based record fetching**. Instead of retrieving every record of a form, a criteria is given so that only the required record is returned. In the Job Application Form workflow, this is written as:

```
jobApplication = Job_Application[ID == input.ID];
```

- The workflow identifies the current record using input.ID, which holds the ID of the record being processed.
- Only the relevant Job Application record, the one whose ID matches input.ID, is retrieved.
- This avoids unnecessarily fetching and processing unrelated records.
- This improves workflow efficiency and reduces unnecessary data processing.

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## Procedure

1. Open the Job Application Form application in Zoho Creator in Edit mode.
2. Navigate to the **Workflow** section and open the **Approvals** tab. The workflow list shows the form Job Application with its workflow.
3. Observe the workflow "HR approval". It runs On review, has 2 Actions, has the status Enabled and was created on 08-Sep-2026 (Figure 1).

[Insert screenshot here: Workflow > Approvals tab showing the HR approval workflow]

*Figure 1: Workflow Section of the Job Application Form Application (Actual Screenshot)*

4. When a Job Application is submitted, the workflow identifies the current record using input.ID.
5. The required record is fetched using the criteria ID == input.ID, so that only this record is retrieved.
6. The workflow checks whether the email field of the record is blank.
7. If the email is available, the notification email is sent to the applicant. If the email is missing, an informational message is displayed.

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## Deluge Code

TheDelugecodeof the workflow is as follows:

```
jobApplication = Job_Application[ID == input.ID];

if(jobApplication.Email.isBlank())
{
    info"Email  address is missing for this application";
}
else
{
    sendmail
    [
        from:zoho.adminuserid
        to :jobApplication.Email
        subject :"Your Job Application is Received"
        message :"<p>Dear Applicant,</p>
        <p>Your job application has been received and is under review.
        Our HRteam will contact you shortly.</p>
        <p>Thank you for applying.</p>"
    ]
}
```

**Explanation:** The first statement fetches the record by its ID. If the email field of that record is blank, an informational message is displayed. Otherwise, an email with the subject "Your Job Application is Received" is sent to the applicant's email address.

## Performance Analysis

Theoptimizedapproachiscompared with a general inefficient approach below.

| Approach | Steps |
|---|---|
| Before (inefficient approach) | Fetch all records, iterate through records, find the required record, process |
| Optimized (used in the workflow) | Fetch the required record using ID, process directly |

The inefficient approach retrieves and processes records that are not needed, whereas the optimized approach reduces unnecessary record retrieval and processing. Figure 2 is a conceptual diagram that illustrates this comparison.




inefficient vs optimized retrieval · two flows

*Figure 2: Inefficient and Optimized Approaches (Conceptual Diagram)*

The "before" code below is an illustrative example only; it was not used in the application.

```
jobApplications = Job_Application[];
efaocrhrecin    jobApplications
{
if(rec.ID==      input.ID)
{
        // ...
}
}
```

No benchmark was performed in this practical, so no measured improvement in execution time is claimed. The comparison is based on the amount of data retrieved and processed.

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## Optimization Techniques

Generalpractices thatcontribute to performance in Zoho Creator applications are given in the table below and shown in Figure 3, a conceptual diagram.

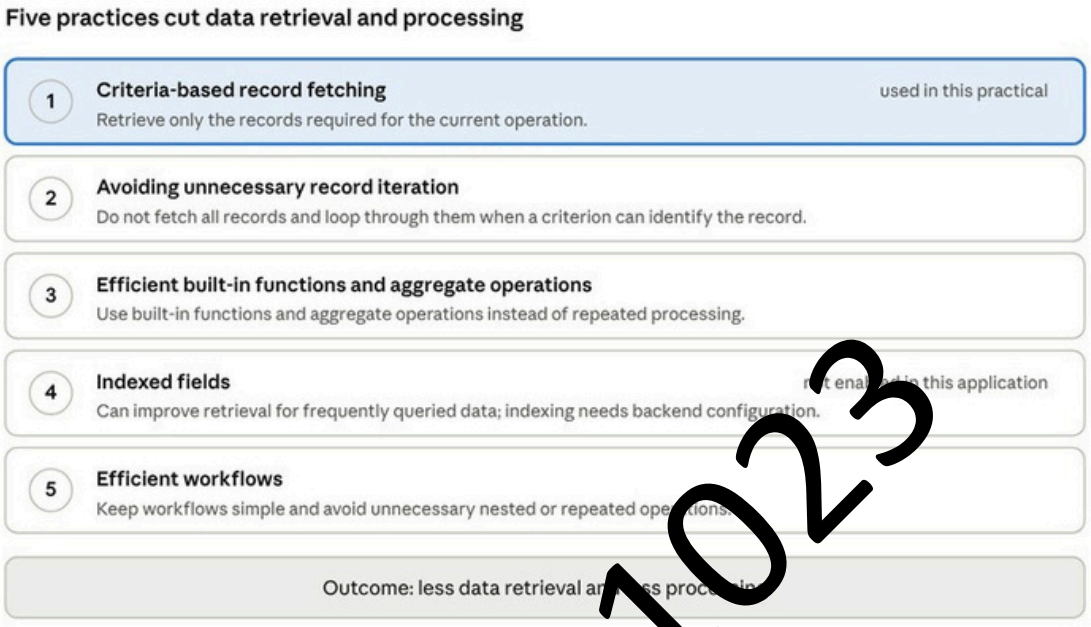


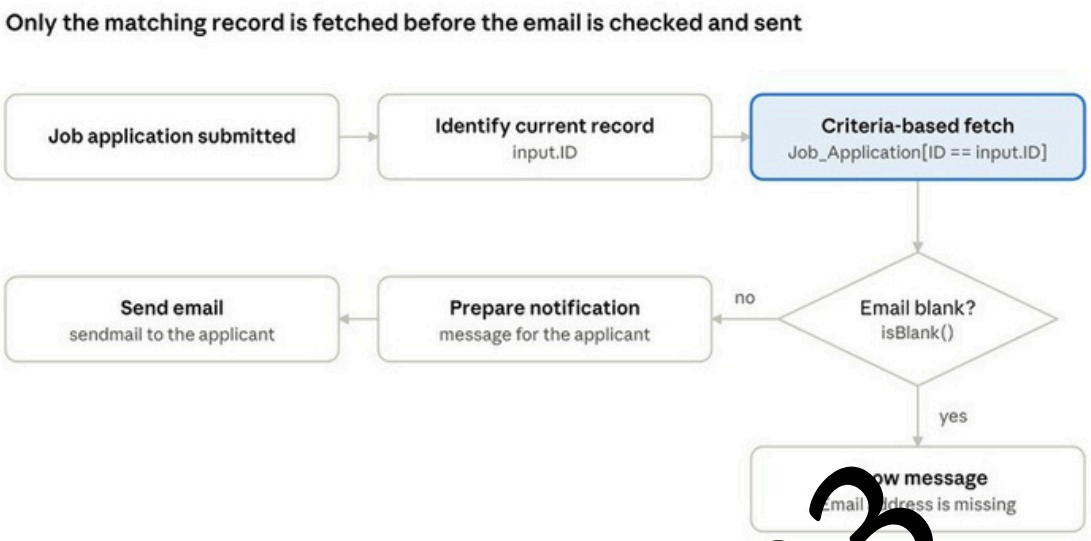

*Figure 3: Performance Optimization Techniques (Conceptual Diagram)*

| Technique | Explanation |
|---|---|
| Criteria-based record fetching | Retrieve only the records required for the current operation. This technique is used in the practical. |
| Avoiding unnecessary record iteration | Do not fetch all records and loop through them when a specific criterion can be used. |
| Efficient built-in functions and aggregate operations | Use built-in functions and aggregate operations instead of unnecessary repeated processing. |
| Indexed fields | Indexed fields can improve retrieval performance for frequently queried data. Indexing may require backend / platform configuration, and it was not enabled or configured in this application. |
| Efficient workflows | Keep workflows simple and avoid unnecessary nested or repeated operations. |

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## Optimized Workflow

Figure4isaconceptualflowchart of the optimized Job Application workflow:




optimized workflow · 6 steps, 1 decision

*Figure 4: Optimized Job Application Workflow (Conceptual Flowchart)*

- **Identify current record:** input.ID gives the ID of the submitted record.
- **Criteria-based record fetch:** Job_Application[ID == input.ID] retrieves only the relevant record.
- **Validate email:** Email.isBlank() checks whether the email field is empty.
- **Prepare notification and send email:** If the email is available, the notification message is prepared and sent using sendmail; otherwise an informational message is displayed.

## Conclusion

Performance optimization is important in low-code applications because efficiency depends not only on the platform but also on how workflows and scripts are written. In this practical, the Job Application Form workflow used criteria-based record retrieval with the expression Job_Application[ID == input.ID], so that only the required record was retrieved and processed.

Targeted record retrieval improves efficiency by avoiding the retrieval of unrelated records and the iteration through them, and efficient workflow design reduces unnecessary processing. Practices such as criteria-based fetching, avoiding unnecessary loops, using built-in operations and keeping workflows simple help applications handle growing amounts of data. These practices contribute to scalable and maintainable Zoho Creator applications. No benchmark was performed, so the benefits are described in terms of reduced data retrieval and processing and not as measured execution times.

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